Stony Brook Institute at Anhui University

安徽大学纽约石溪学院

Chapter 7
Hypothesis test

Hypothesis Testing of Normal Population

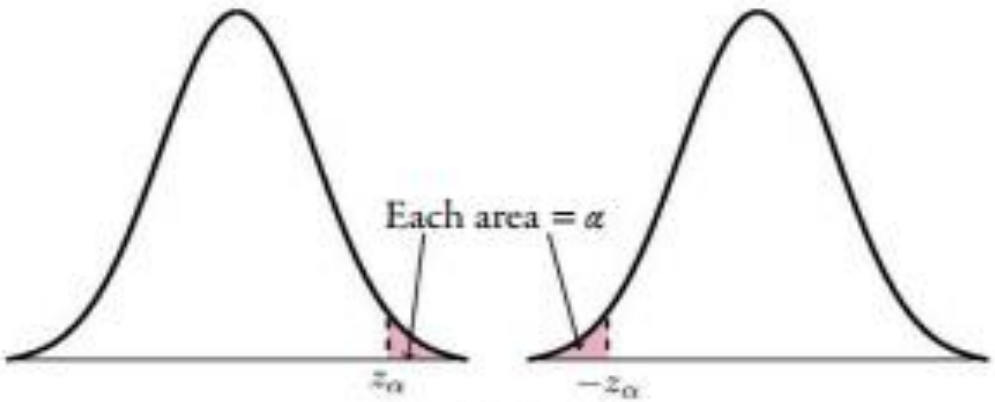
- We will discuss statistical inference for a normal population $\mathrm{N}(\mu, \sigma^2)$. In hypothesis testing, one-sided and two-sided tests are two approaches for evaluating the significance of a parameter using a test statistic.

- **Two side:** $\boldsymbol{H_0 : \mu = \mu_0 \leftrightarrow H_1 : \mu \neq \mu_0}$
- One side: $H_0 : \mu = \mu_0$ or $\mu \geq \mu_0 \leftrightarrow H_1 : \mu < \mu_0$

  $H_0 : \mu = \mu_0$ or $\mu \leq \mu_0 \leftrightarrow H_1 : \mu > \mu_0$

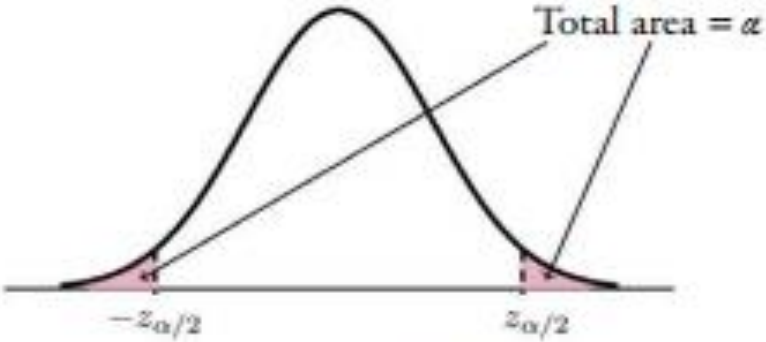
One-sided and two-sided tests

◆Unilateral hypothesis



One-sided tests

◆*Bilateral alternative hypothesis*



Two-sided test

Test about μ for a normal population with known σ

Assuming the variance of a normal population is 49 and the sample mean of 24 values is $\bar{x} = 55.8$, determine whether $\mu \leq 55$ is valid at $\alpha = 0.05$.

Step 1: $H_0: \mu \leq 55, \quad H_1: \mu > 55$

Step 2: $U = \dfrac{\bar{X} - \mu_0}{\sigma / \sqrt{n}} \sim N(0,1)$

Step 3: $z_\alpha = z_{0.05} = 1.64$

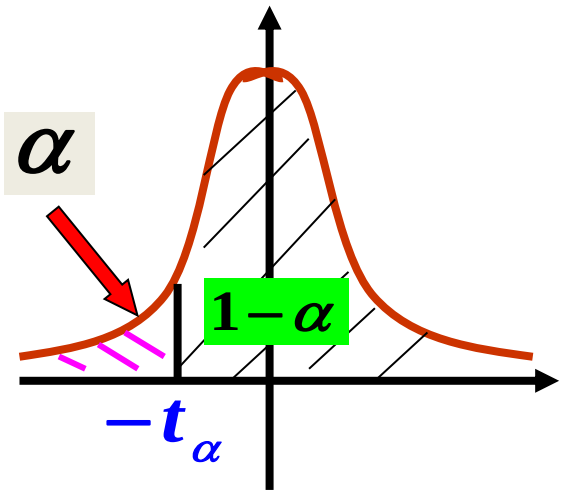
Step4: $\dfrac{\bar{x} - 55}{7/\sqrt{24}} = \dfrac{55.8 - 55}{7/\sqrt{24}} = 0.56 < 1.64$

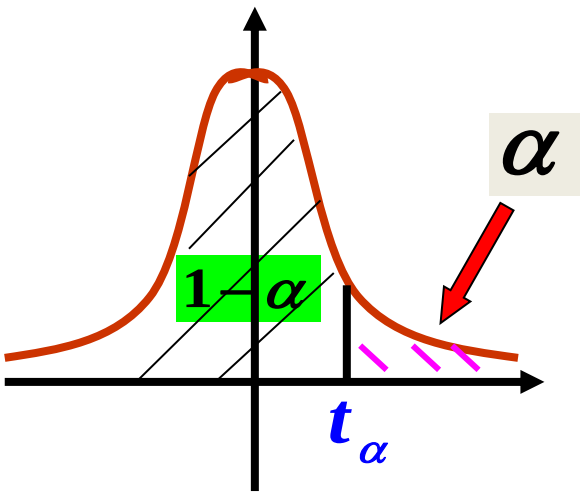
Step5: H_0 cannot be rejected. It can be considered that $\mu \leq 55$ is valid.

Test about μ for a normal population with unknown σ

(2) Unilateral hypothesis

$H_0: \mu = \mu_0, \quad H_1: \mu > \mu_0$ **or** $H_0: \mu = \mu_0, \quad H_1: \mu < \mu_0$





$$\frac{\bar{x}-\mu_0}{s/\sqrt{n}} \le -t_\alpha(n-1)$$

$$\frac{\bar{x}-\mu_0}{s/\sqrt{n}} \ge t_\alpha(n-1)$$

Test about μ for a normal population with unknown σ

例3 某库房要验收大批同类物质，根据以往的经验，这批物质每件的重量服从正态分布。按规定这批物质平均每件重量应为 100 公斤，今抽取10 件，测得其均值 $\bar{x} = 99.6, \quad s^2 = 4.044$

问：能否接受这批物质？$(\alpha = 0.05)$

解: 设 $H_0: \mu = \mu_0 = 100, \quad H_1: \mu \neq \mu_0 = 100$

$\because$ σ^2 未知，所以用 t 检验。

$$\left|\frac{\bar{x} - 100}{s/\sqrt{10}}\right| = \left|\frac{99.6 - 100}{\sqrt{4.044/10}}\right| = 0.629$$

Test about μ for a normal population with unknown σ

Q: A warehouse is inspecting a large batch of similar substances. Based on previous experience, the weight of each item in the batch follows a normal distribution. The standard requirement is that the average weight of each item should be 100 kg. A sample of 10 items is selected, and $\bar{x} = 99.6, \quad s^2 = 4.044$.

Question: Can this batch of substances be accepted at α=0.05?

Step 1: $H_0: \mu = \mu_0 = 100, \quad H_1: \mu \neq \mu_0 = 100$

Step 2: $$T = \frac{\sqrt{n}(\bar{X} - \mu_0)}{S} \sim t(n-1)$$

Step 3: $$t_{\frac{\alpha}{2}}(10-1) = t_{\frac{0.05}{2}}(10-1) = 2.2622$$

Step 4: $$\left|\frac{\bar{x} - 100}{s/\sqrt{10}}\right| = \left|\frac{99.6 - 100}{\sqrt{4.044/10}}\right| = 0.629 \qquad 0.629 < 2.2622$$

Step5: H_0 cannot be rejected.

Test about μ for a normal population with unknown σ

Example 4 The intelligence quotients (IQs) of 5 students from Hefei are: 108, 127, 112, 121, 119. Assuming that the IQ is normally distributed, Does the data indicate that the average IQ of the population in Hefei exceeds 110(α=0.05)?

Step 1: $H_0: \mu = \mu_0$ $H_1: \mu > \mu_0$ $(\mu_0 = 110)$

Step 2:

$$T = \frac{\sqrt{n}(\bar{X} - \mu_0)}{S} \sim t(n-1)$$

Step 3: Rejection region:

T>t_α(n-1) =$t_{0.05}$(5-1)=2.132

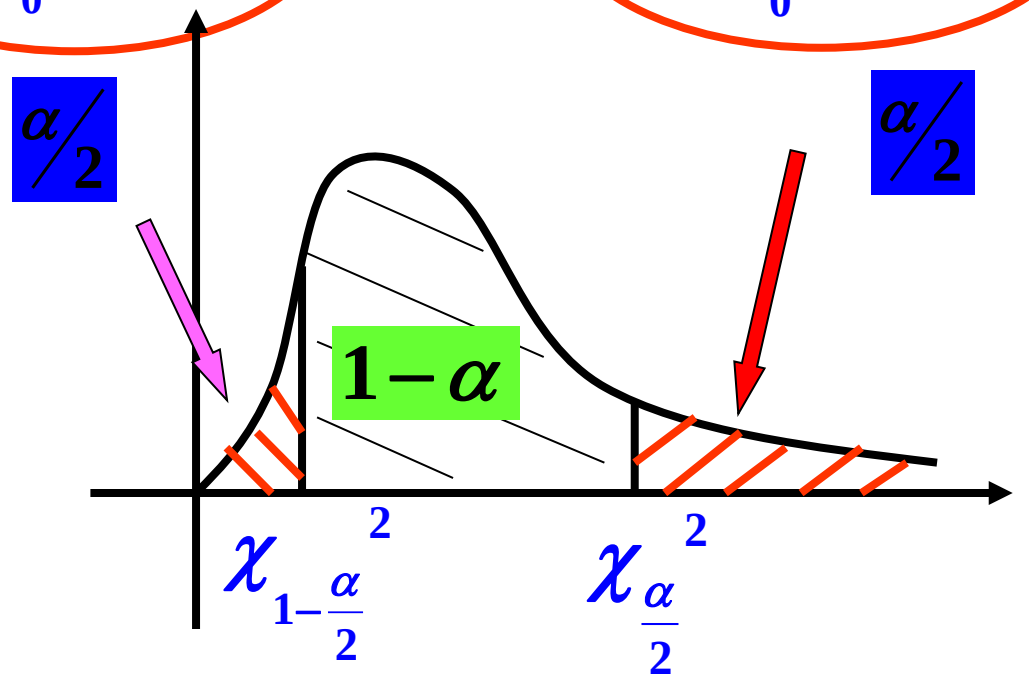
Step 4: Calculate T. n=5, $\bar{x} = 107.4, S^2 = 56.3, S = 7.5$

$$T = \frac{\bar{X}-\mu_0}{S/\sqrt{n}} = \frac{(117.4-110)}{7.5/\sqrt{5}} = 2.206 > 2.132 = t_{0.05}(4)$$

Step 5: Reject H_0 . That is, the data shows that the average IQ of the population in Hefei does not exceed 110.

Test about σ^2 for a normal population with μ unknown

$$\frac{(n-1)s^2}{\sigma_0^2} \le k_1 \quad \text{or} \quad \frac{(n-1)s^2}{\sigma_0^2} \ge k_2$$



Test about σ^2 for a normal population with μ unknown

Step 1: $H_0: \sigma^2 = \sigma_0^2, \quad H_1: \sigma^2 \neq \sigma_0^2$

Step 2:

$$\chi^2 = \frac{(n-1)s^2}{\sigma_0^2} \sim \chi^2(n-1)$$

Step 3:

$$P_{\sigma_0^2}\{(\frac{(n-1)\cdot s^2}{\sigma_0^2} \leq k_1) \cup (\frac{(n-1)\cdot s^2}{\sigma_0^2} \geq k_2)\} = \alpha$$

$$P_{\sigma_0^2}\{(\frac{(n-1)\cdot s^2}{\sigma_0^2} \leq k_1) = \frac{\alpha}{2}, \quad P_{\sigma_0^2}\{(\frac{(n-1)\cdot s^2}{\sigma_0^2} \geq k_2) = \frac{\alpha}{2}$$

$$k_1 = \chi^2_{1-\frac{\alpha}{2}}(n-1), \quad k_2 = \chi^2_{\frac{\alpha}{2}}(n-1)$$

Test about σ^2 for a normal population with μ unknown

Step 4:

$$\frac{(n-1)s^2}{\sigma_0^2} \le \chi^2_{1-\alpha/2}(n-1)$$

$$\frac{(n-1)s^2}{\sigma_0^2} \ge \chi^2_{\alpha/2}(n-1)$$

Step 5: H_0 can/cannot be rejected.

Test about σ^2 for a normal population with μ unknown

Example 7.2. The lifetime(unit: hours) of one certain type of battery obeys normal distribution with $X \sim N(\mu, 5000)$. Now 26 battery are sampled to test the average life of this batch of battery, and the sample variance $S^2 = 9200$. Determine whether $\sigma^2 = 5000$ can be accpted or not? (α=0.02)

Test about σ^2 for a normal population with μ unknown

Solution: 1. $H_0: \ \sigma^2 = 5000 \ \leftrightarrow \ H_1: \sigma^2 \neq 5000$

2. statistic:

$$\chi^2 = \frac{(n-1)S^2}{\sigma_0^{\ 2}} \sim \chi^2(n-1),$$

3. rejection region: $\chi^2 \geq \chi^2_{\alpha/2}$(n-1) or $\chi^2 \leq \chi^2_{1-\alpha/2}$(n-1))

$$\chi^2_{\alpha/2}(n-1) = \chi^2_{0.01}(25) = 44.314, \ \chi^2_{1-\alpha/2}(n-1) = \chi^2_{0.99}(25) = 11.524,$$

4. calculate: n=26, $\alpha = 0.02, \sigma_0^2 = 5000, S^2 = 9200.$

$$\chi^2 = \frac{(n-1)s^2}{\sigma_0^{\ 2}} = \frac{25 \times 9200}{5000} = 46 > 44.314 = \chi^2_{0.01}(25)$$

5. so reject H_0.

Test about σ^2 for a normal population with μ unknown

For a normal population with unknown μ , we use the following procedure for **testing about σ^2**.

Step 1 : $H_0: \sigma^2 = \sigma_0^2 \leftrightarrow H_1: \sigma^2 \neq \sigma_0^2 (\sigma^2 > \sigma_0^2 \; or \; \sigma^2 < \sigma_0^2)$

Step 2: $When \; H_0$ is right, $\chi^2 = \dfrac{(n-1)S^2}{\sigma_0^{\;2}} \sim \chi^2(n-1)$,

Step 3: Rejection region:

$\chi^2 \geq \chi_\alpha^2$(n-1) or $\chi^2 \leq \chi_{1-\alpha/2}^2$(n-1)

($\chi^2 \geq \chi_\alpha^2$(n-1) or $\chi^2 \leq \chi_{1-\alpha}^2$(n-1))

Step 4: Substitute S^2, σ_0^2 and n, calculate χ^2.

if $\chi^2 \geq \chi^2{}_{\alpha/2}$(n-1) or $\chi^2 \leq \chi^2{}_{1-\alpha/2}$(n-1) , reject H_0 .

Test for μ (σ known) --N(μ, σ^2)

	Case (a)	Case (b)	Case (c)
Step 1	$H_0: \mu = \mu_0, H_1: \mu \neq \mu_0$ (2-sided alternative)	$H_0: \mu = \mu_0, H_1: \mu > \mu_0$ (1-sided alternative)	$H_0: \mu = \mu_0, H_1: \mu < \mu_0$ (1-sided alternative)
Step 2	$\alpha = ?$	$\alpha = ?$	$\alpha = ?$
Step 3	$z = \dfrac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$ Rejection region: $\lvert z \rvert \geq z_{\alpha/2}$	$z = \dfrac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$ Rejection region: $z \geq z_{\alpha}$	$z = \dfrac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$ Rejection region: $z \leq -z_{\alpha}$
Step 4	Substitute $\bar{x}, \mu_0, \sigma$, and n Calculate z Decision	Substitute $\bar{x}, \mu_0, \sigma$, and n Calculate z Decision	Substitute $\bar{x}, \mu_0, \sigma$, and n Calculate z Decision

Test for μ (σ unknown) --N(μ, σ^2)

	Case (a)	Case (b)	Case (c)
Step 1	$H_0: \mu = \mu_0, H_1: \mu \neq \mu_0$ (2-sided alternative)	$H_0: \mu = \mu_0, H_1: \mu > \mu_0$ (1-sided alternative)	$H_0: \mu = \mu_0, H_1: \mu < \mu_0$ (1-sided alternative)
Step 2	$\alpha = ?$	$\alpha = ?$	$\alpha = ?$
Step 3	$t = \dfrac{\bar{x} - \mu_0}{s/\sqrt{n}}$ Rejection region: $\lvert t \rvert \geq t_{\alpha/2, n-1}$	$t = \dfrac{\bar{x} - \mu_0}{s/\sqrt{n}}$ Rejection region: $t \geq t_{\alpha, n-1}$	$t = \dfrac{\bar{x} - \mu_0}{s/\sqrt{n}}$ Rejection region: $t \leq -t_{\alpha, n-1}$
Step 4	Substitute $\bar{x}$, μ_0, s, and n Calculate t Decision	Substitute $\bar{x}$, μ_0, s, and n Calculate t Decision	Substitute $\bar{x}$, μ_0, s, and n Calculate t Decision

Test about σ^2 --N(μ, σ^2)

	Case (a)	Case (b)	Case (c)
Step 1	$H_0: \sigma^2 = \sigma_0^2, H_1: \sigma^2 \neq \sigma_0^2$ (2-sided alternative)	$H_0: \sigma^2 = \sigma_0^2, H_1: \sigma^2 > \sigma_0^2$ (1-sided alternative)	$H_0: \sigma^2 = \sigma_0^2, H_1: \sigma^2 < \sigma_0^2$ (1-sided alternative)
Step 2	$\alpha = ?$	$\alpha = ?$	$\alpha = ?$
Step 3	$\chi^2 = \frac{(n-1)s^2}{\sigma_0^2}$ Rejection region: $\chi^2 \geq \chi^2_{\alpha/2, n-1}$ or $\chi^2 \leq \chi^2_{1-\alpha/2, n-1}$	$\chi^2 = \frac{(n-1)s^2}{\sigma_0^2}$ Rejection region: $\chi^2 \geq \chi^2_{\alpha, n-1}$	$\chi^2 = \frac{(n-1)s^2}{\sigma_0^2}$ Rejection region: $\chi^2 \leq \chi^2_{1-\alpha, n-1}$
Step 4	Substitute s^2, σ_0^2 and n Calculate χ^2 Decision	Substitute s^2, σ_0^2 and n Calculate χ^2 Decision	Substitute s^2, σ_0^2 and n Calculate χ^2 Decision

Relationship between CI and test

- The confidence interval and hypothesis test have a close relationship. The confidence interval is the same as the acceptance region in a two-sided test.

- In a two-sided test, the parameter is included in the confidence interval if and only if H_0 is not rejected.
  The parameter is not included in the confidence interval if H_0 is rejected.

Relationship between CI and test

- CI is the same as the acceptance region in a two-sided test.
- The parameter is included in the CI $\Leftrightarrow$ Do not reject H_0 in a two-sided test.
- The parameter is not included in the CI $\Leftrightarrow$ Reject H_0 in a two-sided test.